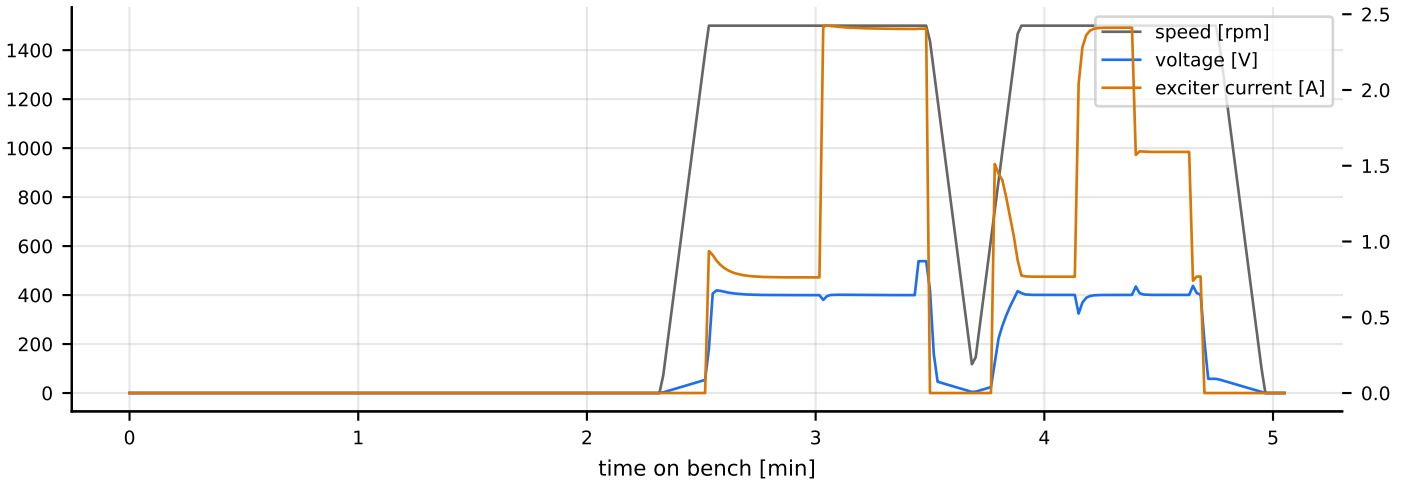
Routine test report

Low-voltage alternator, production test bench (simulation)

Serial	EX-0001-PASS
Frame	medium
Test connection	400 V star, 50 Hz
Test load	640 kW
Test type	routine test
Duration on bench	5.1 min

PASS

Check	Value	Unit	Limits	Result
Stator winding resistance at 20 degC	0.00421	ohm	0.00399 .. 0.00441	OK
Main rotor resistance at 20 degC	1.888	ohm	1.805 .. 1.995	OK
Exciter stator resistance at 20 degC	9.448	ohm	9.025 .. 9.975	OK
Exciter rotor resistance at 20 degC	0.204	ohm	0.199 .. 0.221	OK
Insulation resistance before HV	763.58	Mohm	>= 100.00	OK
Dielectric withstand, peak leakage	22.27	mA	<= 50.00	OK
Phase sequence	correct			OK
Voltage unbalance, no load	0.08879	%	<= 0.500	OK
Exciter field current, no load at Un	0.767	A	0.675 .. 0.825	OK
AVR supply voltage at no load	152.27	V	132.00 .. 168.00	OK
Exciter field current on load at Un	2.405	A	2.145 .. 2.621	OK
Load cut-off voltage / Un	1.347	-	1.250 .. 1.400	OK
Voltage build-up frequency with AVR	27.52	Hz	<= 40.00	OK
No-load voltage with AVR, deviation	0.223	%	-1.000 .. 1.000	OK
Voltage regulation at PF 0.8	0.00226	%	-1.500 .. 1.500	OK
Voltage regulation at PF 1.0	0.00083	%	-1.500 .. 1.500	OK
CT secondary current and polarity	1.001	A	0.900 .. 1.200	OK
Remanent voltage / Un	14.49	%	>= 10.00	OK
PT100 resistance	109.49	ohm	105.00 .. 115.00	OK
PTC resistance	88.56	ohm	20.00 .. 250.00	OK
Space heater resistance	214.41	ohm	196.79 .. 226.41	OK



Simulated unit and bench. Generic limits, not a manufacturer's values.